Benjamin Dixon

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I am an MSc Cognitive Science student at the University of Edinburgh and a Research Scientist at Emory University's Language as a Biomarker Lab. I build computational measures of speech and language as biomarkers of psychosis risk, using large language models, time series methods, and statistical modeling across multiple languages in the AMP SCZ consortium. My interests are computational psychiatry, prediction in language and the brain, neural time series, and information-theoretic approaches to language.

EDUCATION

University of Edinburgh 2026 – 2027
MSc Cognitive Science Edinburgh, UK

Emory University 2021 – 2025
BA Computer Science; BA Psychology & Linguistics · Highest Honors Atlanta, GA, USA

- Thesis: *Automated Assessment of Concrete Language in Clinical High-Risk for Psychosis: A Novel Large Language Model Approach*

PUBLICATIONS

[1] Measuring Cohesion with Self-Attention: Local Focus and Weaker Long-Range Links in Clinical High Risk for Psychosis
Dixon, B. T., et al.
Invited submission, Schizophrenia Bulletin special issue (AMP SCZ baseline analyses). ***In preparation***

[2] Semantic Elements of Interview and Diary Speech Characterize Positive and Negative Symptoms in Clinical High Risk for Psychosis Across Languages: Baseline Findings from the AMP SCZ Observational Study
Bilgrami, Z. R., **Dixon, B. T.**, et al.
Invited submission, Schizophrenia Bulletin special issue (AMP SCZ baseline analyses). ***In preparation***

[3] DIALOG-DeID: Role- and Privacy-Aware Transcription for Clinical Interviews Beyond WER
Oliveira, G. C., **Dixon, B. T.**, et al.
Proceedings of Interspeech 2026, Sydney, Australia.

[4] Large Language Models Can Predict Psychosis Risk from Disruptions in the Flow of Ideas: Results from the AMP SCZ Observational Study
Bilgrami, Z. R., **Dixon, B. T.**, et al.
American College of Neuropsychopharmacology (ACNP) Annual Meeting, 2027. ***Accepted poster***

[5] A Scalable and Extensible Infrastructure for AI-Driven Processing of Clinical Interviews
Tamayo, Z., **Dixon, B. T.**, et al.
ACNP Annual Meeting, 2027. ***Accepted poster***

[6] Altered Pronoun Use in Clinical High Risk for Psychosis Across Languages
Dixon, B. T., et al.
Schizophrenia International Research Society (SIRS) Congress, Florence, Italy, 2026. ***Poster***

[7] Uncovering Universal and Culture-Specific Themes in Psychosis Risk: A Cross-Linguistic Large Language Model Framework
Bilgrami, A., **Dixon, B. T.**, et al.
SIRS Congress, Florence, Italy, 2026. ***Poster***

[8] Reduced Future Orientation and Temporal Switching in Spontaneous Language at Clinical High Risk for Psychosis
Zhang, Y., **Dixon, B. T.**, et al.
SIRS Congress, Florence, Italy, 2026. ***Poster***

[9] Automated Assessment of Language Concreteness in Clinical High-Risk and Healthy Control Speech Using Large Language Models
Zarina, B., **Dixon, B. T.**, Wolff, P.
SIRS Congress, Chicago, IL, USA, 2025. ***Poster***

RESEARCH EXPERIENCE

Language as a Biomarker Lab, Emory University
Sep 2026 – Present
Research Scientist
Atlanta, GA, USA

- Model clinical interviews as **time series** of sentence embeddings and word-level **surprisal**, using **detrended fluctuation analysis** to test whether the long-range temporal structure of speech differs in early **psychosis** (**NumPy**, **scikit-learn**, **Polars**).
- Built attention-pooling **neural network** classifiers (**PyTorch**) that combine sentence-level LLM **embeddings** into one representation per interview, separating clinical high-risk participants from healthy controls (**cross-validation**, **AUROC**).
- Fine-tune LLMs on naturalistic dialogue so that model-based measures of word predictability match human predictability judgments (**model fine-tuning**, **behavioral data**).
- Maintain **GCP** cloud infrastructure; deployed a local LLM assistant on the lab **HPC** cluster for non-technical staff.

Research Software Engineer
Mar 2025 – Aug 2026

- Led design of the speech and language feature set released to the NIMH Data Archive (Release 4) for the AMP SCZ consortium, covering 15,000+ interview transcripts in multiple languages and giving external researchers standardized **biomarkers** of psychosis risk.
- Compared clinical-high-risk and healthy-control groups with **Bayesian mixed-effects models** (brms) in **R**, highlighting group variability.
- Trained linear **decoding models** (ridge, logistic regression) and shallow neural networks on LLM embeddings (Llama 3.3, gpt-oss-120b, DeBERTa) to predict human ratings of concreteness and specificity from transcripts.
- Built cross-linguistic theme discovery with GPU-accelerated **clustering** (HDBSCAN) of sentence embeddings, replacing a method that did not scale to the full dataset.
- Built automated speech-processing pipelines (speaker separation with NVIDIA NeMo and pyannote; transcription with Whisper and Parakeet) on a GPU **HPC cluster**, halving inference time through quantization and batching; processed 1,500 hours of interview audio.

Research Assistant
May 2024 – Mar 2025

- Measured concreteness of speech with **large language models** in clinical high-risk and healthy participants; results presented at SIRS 2025 and in an honors thesis awarded Highest Honors.

Directed Research, Emory University
Sep 2023 – May 2024
Undergraduate Researcher
Atlanta, GA, USA

- Built a question-answering system over technical documentation using retrieval-augmented generation (**RAG**, Haystack) and a fine-tuned Flan-T5-Large model, reducing fabricated answers.

PROJECTS

Word Surprisal and EEG Encoding Models Oct 2026

Independent Project GitHub

- Tested whether language-model **surprisal** predicts **EEG** from 19 people listening to an audiobook, fitting ridge **encoding models** (temporal response functions, **MNE-Python**) with leave-one-run-out cross-validation.
- Built the **signal processing** pipeline (re-referencing, 0.5–8 Hz band-pass filtering, bad-channel interpolation) and aligned EEG with word timings from a separate transcript source.
- Surprisal improved prediction in 18 of 19 participants ($p < 0.001$) beyond sound, word-timing, and word-frequency controls; a **cluster permutation test** localized the effect to a centro-parietal response at about 400 ms (N400).

LifeLine: Plain-Language Guide to Edinburgh Public Services Oct 2026

1st Place, UN Sustainable Development Goals Challenge, AdaHack 2026 GitHub

- Fine-tuned a 1.5B-parameter language model (Qwen2.5, **LoRA**) to rewrite service descriptions in EasyRead plain language for readers with learning disabilities, running offline on a laptop CPU or GPU; team of 3.

Mammography + Chest X-Ray Embedding Fusion Aug 2026

Team Lead, HITI Lab Datathon GitHub

- Collaborated with an 8-person team of engineers and clinicians testing whether combining two imaging models' embeddings predicts 4 chronic diseases better than either alone; fusion improved diabetes and kidney disease screening.
- Compared penalized logistic regression and early-, late-, and gated-fusion neural networks under **nested cross-validation** (**PyTorch**, **scikit-learn**), and evaluated fairness across age, race, and BMI subgroups.

TALKS

College Autism Summit Oct 2024

Developing Autistic Student Leadership (with S. Sabrina and M. Segall) Columbus, OH, USA

TEACHING & MENTORING

Research mentor, three undergraduate research assistants, Emory University 2025 – 2026

Tutor and technical support, data structures and algorithms, Emory University 2023 – 2024

Summer camp instructor, Python and robotics for schoolchildren, Tech Academy, Little Canada, MN 2022

ACTIVITIES & SERVICE

Advanced Team Member, Computer Vision, HumanEd, University of Edinburgh 2026 – Present

Inaugural Cohort, Common Ground Leadership Programme, University of Edinburgh 2026 – 2027

President, Emory Autism Advocacy Organization; grew membership by over 50% 2023 – 2025

Committee Chair, three annual Autism Celebration Months, including a research symposium 2023 – 2025

Co-lead, campus sensory-friendly spaces initiative, Emory University 2023 – 2025

HONORS & AWARDS

1st Place, UN Sustainable Development Goals Challenge, AdaHack 2026, University of Edinburgh 2026

Highest Honors, Emory College of Arts and Sciences (honors thesis) 2025

Highest Excellence in Psychology & Linguistics, Emory University 2025

Member, Psi Chi International Honor Society in Psychology 2024

TECHNICAL SKILLS

Methods	EEG analysis, signal processing, time series analysis, encoding/decoding models, Bayesian and mixed-effects modeling, machine learning, clustering, NLP, LLM fine-tuning
Programming	Python (MNE, NumPy, pandas, Polars, scikit-learn, PyTorch, HuggingFace, vLLM), R (brms, GAMLSS), SQL, Bash, Java, C/C++, JavaScript
Infrastructure	HPC, Slurm, GPU computing, Docker, Git, GCP, AWS (SageMaker, S3)